EX. No.3.1. CREATE 3D MODEL FROM ORTHOGRAPHIC PROJECTION

AIM:

To create a 3D model from the given orthographic projection.

REQUIREMENTS:

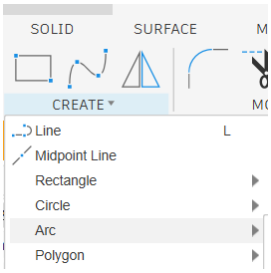
1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster

PROCEDURE:

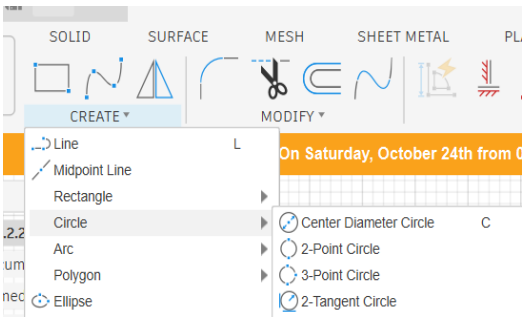
1. Open Autodesk Fusion 360 and create a new design. Set the units to millimetres (mm).

2. Select the Front Plane (XZ Plane) and click Create Sketch.

3. Draw the main front-view profile using the Line and Arc tools. Create the rounded left end with an outer radius of R16 mm and extend the rectangular portion according to the given dimensions.

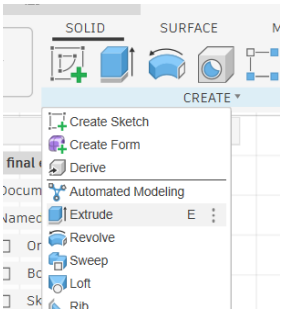


4. Draw the circular hole at the centre of the rounded portion using the Circle tool and set its diameter to Ø16 mm.

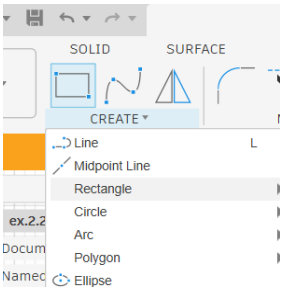


5. Complete the front profile using the given dimensions: 36 mm, 32 mm and the overall height of 32 mm. Use the Sketch Dimension (D) tool to set accurate measurements.

6. Finish the sketch and use the Extrude (E) command to give the model a width of 40 mm, as shown in the right view.



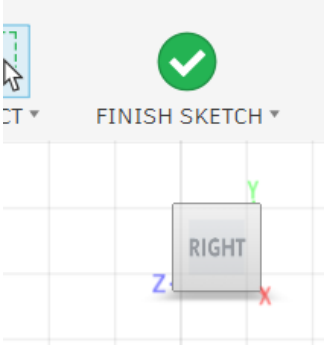
7. Select the Top Face and create a new sketch. Draw a rectangular slot of 18 mm width, leaving 11 mm thickness on both sides (11 + 18 + 11 = 40 mm).



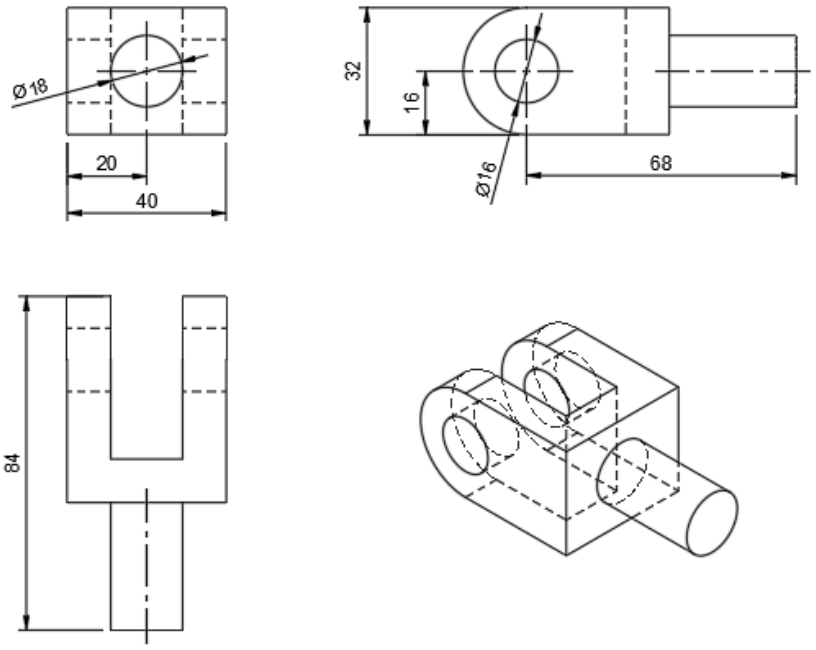
8. Use Extrude → Cut to remove the material and form the slot as shown in the top view. Set its length according to the drawing, stopping at the shoulder.

9. Check the circular hole and remaining dimensions. The right view indicates a hole of Ø18 mm, whereas the front view indicates Ø16 mm. Verify the diameter before finalising the hole.

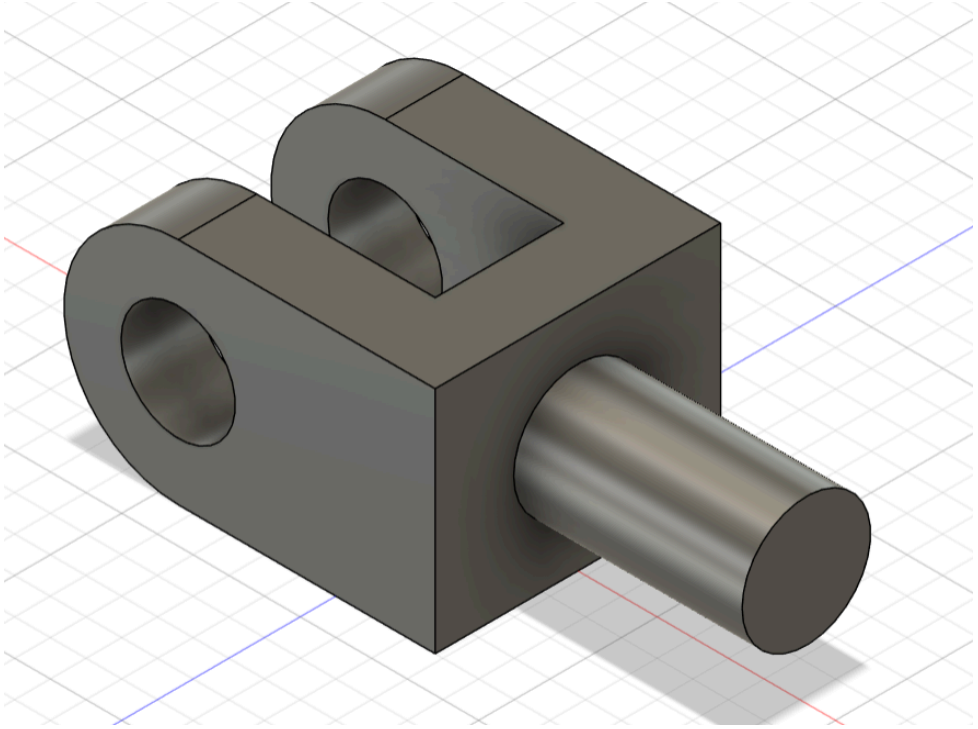
10. Click Finish Sketch, inspect the completed 3D model from the front, top and right sides, and save the design.



INPUT:



OUTPUT:



RESULT:

Thus, the 3D model is created from orthographic projection.